

Project Abstract: Next-Gen Vehicle Rental Aggregator Platform

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Date: September 4, 2026

1. Executive Summary

This project outlines the development of a highly scalable, on-demand aggregator platform for local vehicle rentals (bikes and cars). Functioning as a "Swiggy for rentals" (without doorstep delivery), the platform connects customers directly with verified local rental vendors. It solves the critical pain point of digital visibility for small rental businesses while providing users with a unified, transparent, and advance-booking interface.

2. System Architecture

To ensure a seamless two-sided marketplace, the ecosystem is divided into three distinct portals:

- **Customer App:** A mobile application for users to discover nearby vehicles, complete digital KYC, make advance bookings, and execute secure payments.
- **Vendor App / Dashboard:** A dedicated portal for rental owners to manage their fleet inventory, update real-time availability, and track earnings.
- **Admin Panel:** A centralized control system for platform administration, vendor onboarding, commission management, and dispute resolution.

3. Core Features

Feature Category	Description
Exclusive Partner Mapping	Integration with Google Maps to display only officially onboarded rental hubs, ensuring quality control and exclusive partnerships.
Dynamic Filtering	Users can sort available vehicles by type (scooter, cruiser, hatchback, SUV), transmission, fuel type, and pricing.

Feature Category	Description
Transparent Checkout	Clear price breakdowns detailing base rent, refundable security deposits, platform fees, and taxes prior to payment.
Vendor Fleet Sync	One-tap toggles for vendors to instantly mark vehicles as "Booked," "Available," or "In Maintenance."

4. Smart AI Integrations

To distinguish the platform from standard directories, advanced AI capabilities are integrated to automate heavy lifting and reduce manual verification:

- **Automated KYC (OCR):** Leveraging optical character recognition (e.g., Google Cloud Vision API) to automatically extract, parse, and verify details (Name, License Number, Expiration) from uploaded driver's licenses.
- **Automated Damage Detection:** Utilizing computer vision models (such as YOLO architecture) during the vehicle pickup phase. Customers capture a 360-degree video/photos, and the system automatically logs pre-existing dents or scratches to prevent post-rental disputes.

5. Technology Stack

The platform will be built using modern, scalable technologies tailored for rapid deployment and high performance.

Frontend / Mobile Apps	: Google Studio (Project IDX) / Flutter (Dart) or Native Android
Backend & Database	: Google Firebase (Firestore NoSQL, Cloud Functions)
Authentication	: Firebase Auth (OTP & Google Sign-In)
Mapping & Location	: Google Maps SDK
AI Services	: Google Cloud Vision API

6. Monetization Strategy

- **Commission Model:** A flat percentage fee (e.g., 10-15%) charged to the rental vendor for every successful booking facilitated by the platform.
- **Promoted Listings:** An optional premium tier allowing rental owners to boost their visibility and appear at the top of local search results for a fee.